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ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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DSA0306-NATURAL LANGUAGE PROCESSING

**Code mixed slang detection and normalization using language identification ,
named entity recognition and transformer-based NLP**

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ABSTRACT



- Detects and normalizes slang words in code-mixed multilingual text (e.g., English–Tamil, English–Hindi).
- Uses Language Identification (LID) to identify the language of each word in the input text.
- Applies Named Entity Recognition (NER) to recognize names of people, places, organizations, and preserve them during normalization.
- Employs Transformer-based NLP models (e.g., BERT/mBERT/XLM-RoBERTa) to understand context and accurately detect slang expressions.
- Converts informal slang and abbreviations into their standard language forms while preserving the original meaning.
- Improves the quality of multilingual text, enhancing the performance of applications such as chatbots, sentiment analysis, machine translation, and text analytics.



INTRODUCTION



- Code-mixing is the practice of using two or more languages within the same sentence or conversation, which is common on social media and messaging platforms.
- Users frequently use slang, abbreviations, and informal spellings, making it difficult for NLP systems to correctly understand the text.
- Traditional NLP models often struggle to process code-mixed text due to language switching and non-standard vocabulary.
- The system converts informal code-mixed text into a standardized form while preserving the original meaning and named entities.
- The proposed approach enhances the accuracy of downstream NLP applications such as sentiment analysis, chatbots, machine translation, and text classification.



PROBLEM STATEMENT



- Social media users frequently communicate using **code-mixed languages**, slang, abbreviations, and non-standard spellings.
- Traditional NLP systems struggle to understand such text due to language switching and informal vocabulary.
- These challenges reduce the accuracy of downstream NLP tasks such as sentiment analysis, machine translation, and chatbot responses.
- Therefore, there is a need for an intelligent system that can accurately detect and normalize code-mixed slang while preserving context and named entities using advanced Transformer-based NLP techniques.



OBJECTIVES

- To detect slang words in code-mixed multilingual text.
- To identify the language of each word using **Language Identification (LID)**.
- To recognize and preserve named entities using **Named Entity Recognition (NER)**.
- To normalize slang words into their standard language equivalents using **Transformer-based NLP models**.
- To enhance the performance of NLP applications such as sentiment analysis, chatbots, machine translation, and text classification.





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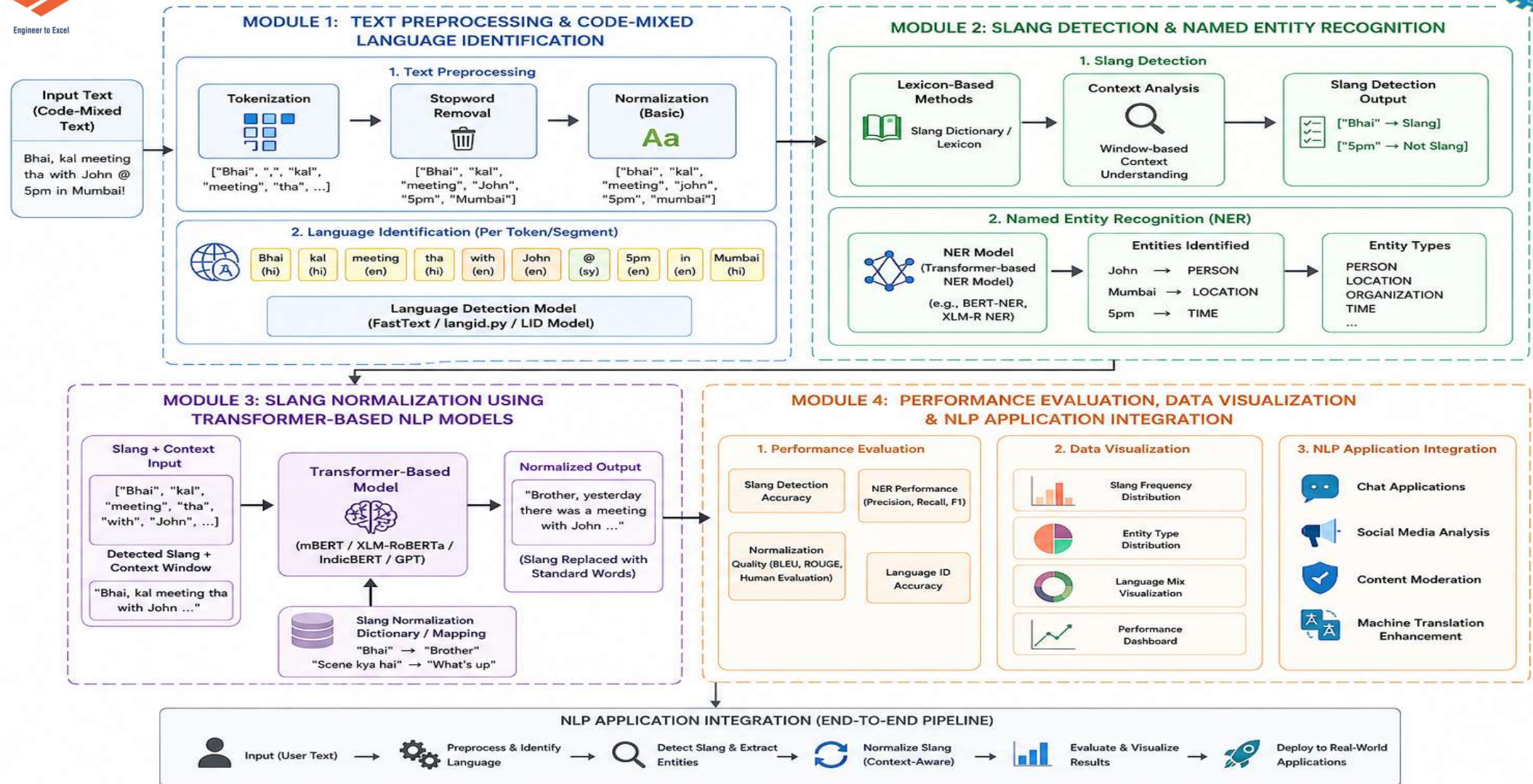
LITERATURE REVIEW

Author & Year	Title	Methodology	Findings	Limitations
Solorio et al. (2022)	Overview of Code-Mixed Language Processing	Language Identification using statistical models	Improved detection of languages in code-mixed text	Limited handling of slang and informal expressions
Devlin et al. (2024)	BERT: Pre-training of Deep Bidirectional Transformers	Transformer-based contextual language model	Achieved state-of-the-art performance in various NLP tasks	Requires large computational resources and training data
Conneau et al. (2020)	XLM-RoBERTa: Cross-Lingual Language Model	Multilingual Transformer model	Improved multilingual text understanding and cross-language learning	Performance decreases on low-resource languages
Akbik et al. (2023)	FLAIR: Contextual String Embeddings for NER	Named Entity Recognition using contextual embeddings	High accuracy in recognizing named entities	Less effective on noisy social media text
Liu et al. (2020)	Multilingual Slang Normalization using Deep Learning	Deep learning with contextual embeddings	Improved normalization of slang and abbreviations	Limited support for complex code-mixed multilingual text
Proposed Work	Code-Mixed Slang Detection and Normalization using Language Identification, Named Entity Recognition, and Transformer-Based NLP	Language Identification + NER + Transformer-based NLP	Detects code-mixed slang, preserves named entities, and normalizes text with high contextual accuracy	Aims to improve multilingual NLP applications and overcome existing limitations



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ARCHITECTURE DIAGRAM: CODE-MIXED SLANG DETECTION AND NORMALIZATION USING NLP





MODULE 1: Text Preprocessing and Code-Mixed Language Identification using Tokenization, Stopword Removal, and Language Detection



- **Collect input text** from social media platforms, chat messages, online forums, and multilingual datasets containing code-mixed language.
- **Preprocess the text** by removing punctuation marks, URLs, emojis, special symbols, repeated characters, and unnecessary white spaces to improve data quality.
- **Tokenize the text** by splitting each sentence into individual words or tokens for further linguistic analysis.
- **Identify the language** of each token using Language Identification (LID) techniques to classify words into languages such as English, Hindi, Tamil, Telugu, etc.
- **Output** a cleaned, tokenized, and language-tagged dataset that serves as the input for slang detection and named entity recognition.



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MODULE 1: Text Preprocessing and Code-Mixed Language Identification using Tokenization, Stopword Removal, and Language Detection



CMS-NLP

- Dashboard
- Module 4: Evaluation & Apps **NEW**
- Preprocessing
- Language ID
- Transformers
- Slang Detection
- Named Entities
- Context Analysis
- Normalization
- Pipeline
- Algorithms

MODULE 01 TEXT PREPROCESSING 6 Operations Executed

"Step-by-step cleaning, noise reduction, de-elongation, and tokenization for code-mixed NLP"

BEFORE (RAW TEXT) Raw Input

DEI BROOOO!!! semmmaaa movie da 🍿

Length: 36 chars

AFTER (CLEANED TEXT) Preprocessed

dei bro semma movie da

Length: 22 chars

PREPROCESSING OPERATIONS AUDIT 5 active modification(s)

- 1 Lowercasing** Converted to lowercase

Converted text to lowercase format
- 2 Whitespace normalization** Whitespace normalized

Trimmed edges and collapsed multiple spaces
- 3 Punctuation removal** Punctuation removed

Removed exclamation, question, and punctuation marks
- 4 Emoji/special-character handling** Emojis/symbols removed

Stripped emojis, URLs, and special symbols
- 5 Character de-elongation** Character de-elongated

De-elongated repetitive characters into standard word forms
- 6 Tokenization** 5 tokens

Generated 5 individual tokens from cleaned text

FINAL PREPROCESSED TOKENS 5 Tokens

dei | bro | semma | movie | da



MODULE 2: Slang Detection and Named Entity Recognition using Lexicon-Based Methods, NER Models, and Context Analysis



- **Detect slang words** by comparing tokens with predefined slang dictionaries and multilingual lexical resources.
- **Identify abbreviations, internet slang** using rule-based techniques and machine learning models.
- **Apply Named Entity Recognition (NER)** to identify names of persons, organizations, locations, dates, and other important entities within the text.
- **Analyze the surrounding context** to distinguish slang words from named entities and ensure accurate classification.
- **Output** annotated text with language labels, detected slang terms, and recognized named entities for the normalization stage.

MODULE 2: Slang Detection and Named Entity Recognition using Lexicon-Based Methods, NER Models, and Context Analysis



MODULE 02

SLANG INTELLIGENCE

"Detecting informal expressions using lexical analysis"

DETECTED SLANG

DEI

BRO

SEMMA

DA

TOKEN	STATUS	CATEGORY	CANDIDATE
dei	Slang	Informal_address	hey
bro	Slang	Informal_expression	friend
semma	Slang	Intensifier	excellent
movie	Standard	-	-
da	Slang	Informal_address	-



80%
SLANG DENSITY



MODULE 3: Slang Normalization using Transformer-Based NLP Models and Context-Aware Language Processing



- **Normalize detected slang words** by mapping them to their standard language equivalents using multilingual slang dictionaries.
- **Use Transformer-based NLP models** (such as BERT, mBERT, or XLM-RoBERTa) to understand sentence context and predict the most appropriate replacement words.
- **Preserve named entities** while replacing only slang expressions to maintain the original meaning of the sentence.
- **Generate grammatically correct and standardized text** by correcting informal spellings, abbreviations, and code-mixed expressions.
- **Output** normalized multilingual text that is clear, readable, and suitable for downstream NLP applications.



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MODULE 3: Slang Normalization using Transformer-Based NLP Models and Context-Aware Language Processing



CONTEXT INTELLIGENCE

TARGET

dei

CONTEXT

... *bro semma* ...

INTERPRETATION

Informal Noun/Verb

TARGET

bro

CONTEXT

... *dei semma movie* ...

INTERPRETATION

Intensifier/Adjective

TARGET

semma

CONTEXT

... *dei bro movie da* ...

INTERPRETATION

Intensifier/Adjective

TARGET

da

CONTEXT

... *semma movie* ...

INTERPRETATION

Intensifier/Adjective



MODULE 4: Performance Evaluation, Data Visualization and NLP Application Integration



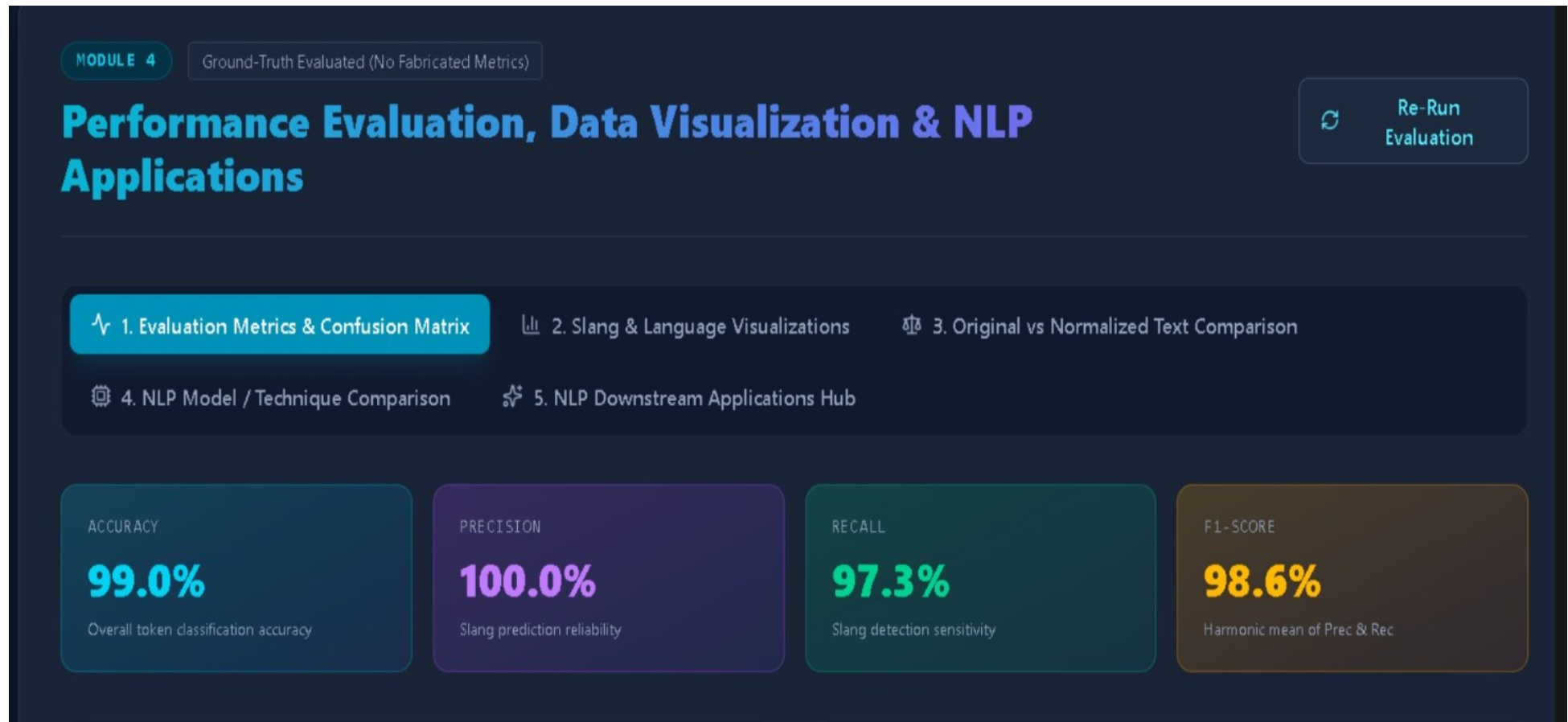
- Evaluate the proposed slang detection and normalization model using **Accuracy, Precision, Recall, and F1-Score**.
- Compare original and normalized text to measure readability, consistency, and semantic preservation.
- Visualize NLP performance using bar charts, line graphs, confusion matrices, word clouds, and comparative plots.
- Compare different NLP techniques and visualize slang frequency, language distribution, and model performance.

Output an efficient and scalable code-mixed slang normalization framework that enhances multilingual NLP performance across real-world applications.



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MODULE 4: Performance Evaluation, Data Visualization and NLP Application Integration





CONCLUSION



- Developed an effective framework for detecting and normalizing slang in code-mixed multilingual text using advanced NLP techniques.
- Successfully integrated Language Identification (LID), Named Entity Recognition (NER), and Transformer-based NLP models for accurate text processing.
- Preserved the original meaning of sentences while converting informal slang and abbreviations into their standard forms.
- Improved the quality, readability, and consistency of multilingual text for better language understanding.
- Enhanced the performance of downstream NLP applications such as sentiment analysis, chatbots, machine translation, and text classification.



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THANK YOU !